Three facts about the empirical distribution function

Abstract

We collect an elementary property of the empirical distribution function of a sample, its almost sure uniform convergence, and the limit law of the Kolmogorov–Smirnov statistic.

1 Setting

Definition 1.1. For $n \geq 1$ and $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, the *empirical distribution function* is $F_n(t) = \frac{1}{n}\#\{i : x_i \leq t\}$ for $t \in \mathbb{R}$.

When $X_1, X_2, \ldots$ are random variables, F_n denotes the empirical distribution function of $(X_1, \ldots, X_n)$, and $D_n = \sup_{t \in \mathbb{R}} |F_n(t) - F(t)|$, where F is the distribution function of X_1.

2 Results

Lemma 2.1. *For every $x \in \mathbb{R}^n$, F_n is nondecreasing, and $nF_n(t) \in \{0, 1, \ldots, n\}$ for every $t \in \mathbb{R}$.*

Proof. If $s \leq t$, every i with $x_i \leq s$ also satisfies $x_i \leq t$, so $F_n(s) \leq F_n(t)$. The number $nF_n(t)$ counts a subset of $\{1, \ldots, n\}$. □

Proposition 2.2. *If $X_1, X_2, \ldots$ are independent and identically distributed, then $D_n \to 0$ almost surely.*

Proof. This is the Glivenko–Cantelli theorem. □

Theorem 2.3. *If $X_1, X_2, \ldots$ are independent and identically distributed with a continuous distribution function F, then $\sqrt{n}\, D_n$ converges in distribution to $\sup_{0 \leq u \leq 1} |B(u)|$, where B is a Brownian bridge on $[0, 1]$. The limit has the distribution function $K(z) = 1 - 2\sum_{k \geq 1} (-1)^{k-1} e^{-2k^2 z^2}$ for $z > 0$.*

Proof. The convergence is Donsker's invariance principle for the empirical process [1], applied to the supremum, which is a continuous functional. The form of K is due to Kolmogorov [2]. □

References

[1] M. D. Donsker. Justification and extension of Doob's heuristic approach to the Kolmogorov–Smirnov theorems. *Annals of Mathematical Statistics*, 23:277–281, 1952.

[2] A. N. Kolmogorov. Sulla determinazione empirica di una legge di distribuzione. *Giornale dell'Istituto Italiano degli Attuari*, 4:83–91, 1933.